

Step 13 - Fraud Detection Lab

Phase E · Structuring · Velocity · STR Candidates · ~20 min

Goal: The capstone lab - fraud and AML detection rules running on your Silver data: structuring detection, velocity anomalies, and an alerts table in Gold. Artifact: 07_fraud_detection.ipynb

LEARN

Rule-based AML detection is the production baseline at every bank (ML augments it; rules remain for explainability - regulators require reasons).

Rule	Pattern	FINTRAC relevance
Structuring	Multiple transactions just below \$10,000 within a window	Deliberate threshold avoidance - STR territory
Velocity	Unusually many transactions in a short window	Account takeover / mule behaviour
Large single	Any transaction $\geq$ \$10,000	LCTR (already flagged in Silver)

LAB — notebook 07_fraud_detection (Serverless)

Cell 1 — Setup:

```
import pyspark.sql.functions as F
from pyspark.sql.window import Window
VOL = "/Volumes/workspace/default/maplebank"
SILVER, GOLD = f"{VOL}/silver", f"{VOL}/gold"
df = spark.read.format("delta").load(f"{SILVER}/transactions") \
    .filter(~F.col("transaction_id").rlike("^(LATE_|TXN_NEW_)")) # exclude Step 7 test rows
```

Cell 2 — Rule 1, Structuring (2+ e-Transfers of \$9,000-\$9,999 within 7 days):

```
days = lambda n: n * 86400
w7 = Window.partitionBy("customer_id") \
    .orderBy(F.col("transaction_timestamp").cast("long")) \
    .rangeBetween(-days(7), 0)

df_structuring = (df
    .filter((F.col("amount_cad") >= 9000) & (F.col("amount_cad") < 10000))
    .filter(F.col("transaction_type") == "INTERAC_ETRANSFER")
    .withColumn("near_threshold_7d", F.count("*").over(w7))
    .filter(F.col("near_threshold_7d") >= 2)
    .withColumn("rule", F.lit("STRUCTURING"))
    .withColumn("severity", F.lit("HIGH")))
print(f"Structuring hits: {df_structuring.count()}")
```

Cell 3 — Rule 2, Velocity (more than 5 transactions in 1 hour):

```
w1h = Window.partitionBy("customer_id") \
    .orderBy(F.col("transaction_timestamp").cast("long")) \
    .rangeBetween(-3600, 0)

df_velocity = (df
    .withColumn("txns_1h", F.count("*").over(w1h))
    .filter(F.col("txns_1h") > 5)
    .withColumn("rule", F.lit("VELOCITY"))
    .withColumn("severity", F.lit("MEDIUM")))
print(f"Velocity hits: {df_velocity.count()}")
```

Cell 4 — Rule 3, LCTR (the Silver flag, formalized):

```
df_lctr = (df.filter(F.col("fintrac_flag") == "LCTR_REPORTABLE")
    .withColumn("rule", F.lit("LCTR_LARGE_TXN"))
    .withColumn("severity", F.lit("REPORT")))
```

Cell 5 — Combine into the alerts table + enrich with MASKED customer data:

```
cols = ["transaction_id", "customer_id", "account_id", "transaction_timestamp",
    "amount_cad", "transaction_type", "rule", "severity"]
df_alerts = df_structuring.select(cols) \
    .unionByName(df_velocity.select(cols)) \
    .unionByName(df_lctr.select(cols))

df_cust = spark.read.format("delta").load(f"{SILVER}/customers") \
    .select("customer_id", "sin_masked", "city", "province", "postal_code_fsa")
df_alerts_enriched = df_alerts.join(df_cust, "customer_id", "left") \
    .withColumn("alert_date", F.current_date())

df_alerts_enriched.write.format("delta").mode("overwrite") \
    .save(f"{GOLD}/fraud_alerts")

df_alerts_enriched.groupBy("rule", "severity").count().show()
df_alerts_enriched.orderBy(F.col("amount_cad").desc()).show(10, truncate=False)
```

Cell 6 — The STR candidates view (structuring + velocity, the human-review queue):

```
spark.read.format("delta").load(f"{GOLD}/fraud_alerts") \
    .filter(F.col("rule").isin("STRUCTURING", "VELOCITY")) \
    .groupBy("customer_id", "sin_masked", "city", "province") \
    .agg(F.count("*").alias("alert_count"),
        F.collect_set("rule").alias("rules_triggered"),
        F.round(F.sum("amount_cad"), 2).alias("total_flagged_cad")) \
    .orderBy(F.col("alert_count").desc()).show(10, truncate=False)
```

COMMIT + CHECKPOINT

Export to databricks/. Pass criteria: counts per rule print; alerts table lands in Gold; STR-candidate view ranks customers. Note in the notebook (and say in interviews): LCTR is automatic reporting; STR requires human investigation - the pipeline produces CANDIDATES, an analyst decides. That nuance reads senior.

Interview line: "My fraud layer is rule-based for explainability - structuring via a 7-day rangeBetween window on just-below-threshold e-Transfers, velocity via a 1-hour window, LCTR straight from the Silver flag - all landing in a Gold alerts table joined only to masked customer attributes."